Farming

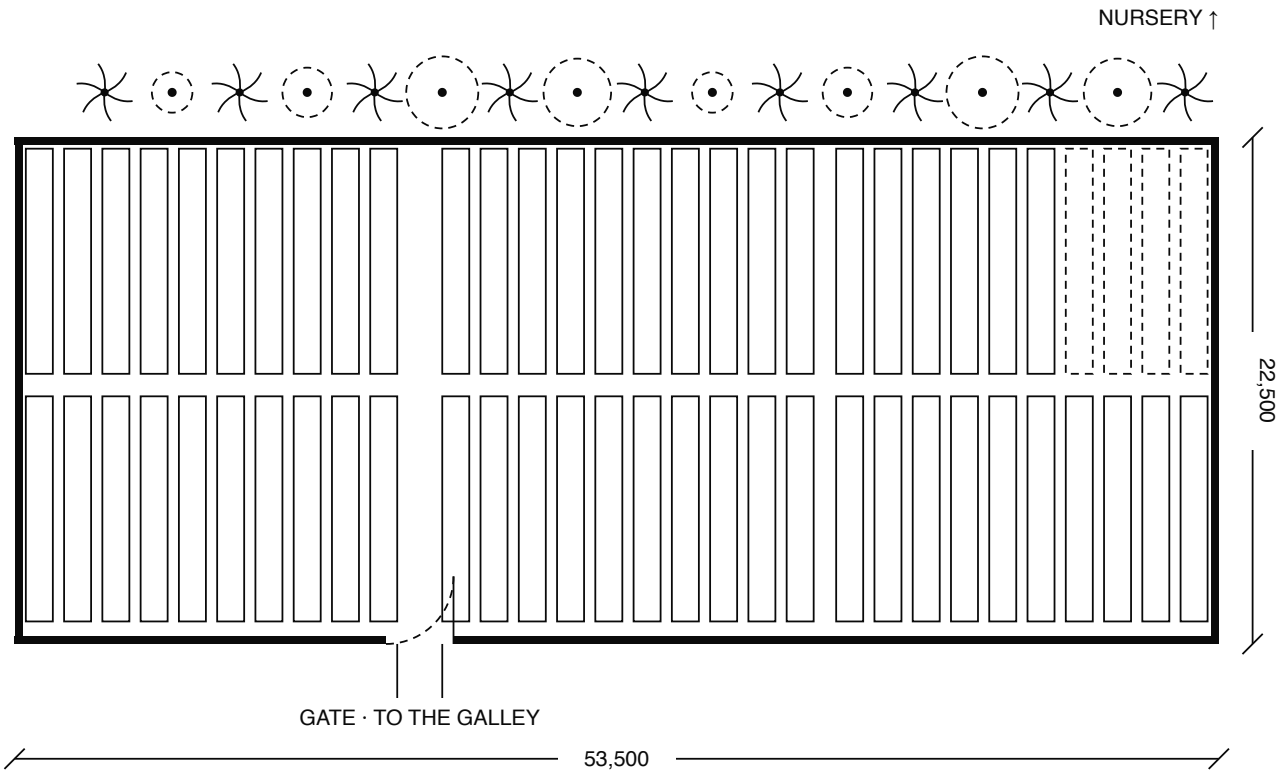
Execution plan

How to start the vegetable field that feeds the galley and sells its surplus: the seasons, the beds, the water, the crops and the first month, step by step.

Status: plan of 2026-10-31. The field, its soil and its water are not yet measured; its size follows the provisional master plan and the field's generated plan (D-050).

Kinyarwanda: soma iyi gahunda mu Kinyarwanda · Print: this plan as a PDF · The beds, crops and costs drawn: the field page · Its other pages: materials, water

At a glance



What	Size
The vegetable zone	3456 m², next to the galley (master plan)
Beds	1.2 m wide, raised 20 cm, 10 m long: 60 beds to start, in 6 blocks of 10 (D-050), 672 m² of them planted [E]
Paths	0.5 m between beds, 1 m between blocks of beds

What	Size
Water	drip lines, 2 per bed, fed from the tank; 2.8 to 3.6 m³ a day in the driest months [E]
What the galley needs	about 38.5 kg of vegetables in an average week, 89.8 kg on a full house [A]; the first beds cover an average week, and a full house needs 105 (the field page); the rest is sold

A full house every day is 25.5 people at the galley's table, 22 guests and the team, about 89.8 kg of vegetables a week, and would take 105 beds. The field starts smaller: the two rows of blocks nearest the gate, 60 beds, 56 of them planted, about 672 m² of growing area with a quarter more than the plate needs for pests and a failed planting. They give about 51.4 kg a week: 57 % of a full house and more than an average week (38.5 kg), whose surplus is sold. On the busiest weeks the galley buys the rest; the other two rows follow the bookings. Behind the field a row of bananas and fruit trees (avocado, mango, lemon, papaya) gives the table its fruit.

Crop	Per guest a day	Full house, a week	Average week	Beds at the start	Full ho
Tomato	80 g	14.3 kg	6.1 kg	9	
Potato	60 g	10.7 kg	4.6 kg	8	
Onion	50 g	8.9 kg	3.8 kg	4	
Cabbage	50 g	8.9 kg	3.8 kg	4	
Carrot	40 g	7.1 kg	3.1 kg	4	
Kale (sukuma)	40 g	7.1 kg	3.1 kg	3	
French beans	35 g	6.2 kg	2.7 kg	8	
Amaranth (dodo)	30 g	5.4 kg	2.3 kg	2	
Cucumber	30 g	5.4 kg	2.3 kg	2	
Spinach and chard	30 g	5.4 kg	2.3 kg	3	
Sweet pepper	25 g	4.5 kg	1.9 kg	3	
Eggplant	25 g	4.5 kg	1.9 kg	4	
All vegetables	**503 g**	**89.8 kg**	**38.5 kg**	**56 of 60**	

Crop	Per guest a day	Full house, a week	Average week	Beds at the start	Full ho
Herbs	5 g	0.9 kg	0.4 kg	1	
Chilli	3 g	0.5 kg	0.2 kg	1	
All vegetables	**503 g**	**89.8 kg**	**38.5 kg**	**56 of 60**	

Grams on the plate are assumed until the cook writes the menu. Eggs come from the hens; events buy what the field can't give.

The seasons

Season	Months	What the field does
A	September to February	Raise seedlings for 3 to 8 weeks, transplant with the rains, drip through the January-February lull
B	March to June	Transplant in February and March, raised beds for the April rains, drip again from late May
C	July to September	Drip only: about 4 to 5 mm of water a day evaporates; plant only what the proven water can carry

Seasons [V LOC-53]; rain and evaporation at Gashora [E LOC-34, LOC-54]; nursery times [V CRP-68].

The crops

Crop	Spacing (cm)	First harvest	Yield	Watch for
Tomato	100 × 50	60 to 90 days after transplanting	about 8 t/ha nationally [V CRP-17]	Tuta absoluta, late blight, bacterial wilt
Onion	20 × 10	90 to 150 days	up to 60 t/ha (variety claims)	thrips, downy mildew
Cabbage	60 × 50 to 60	60 to 105 days	25 to 30 t/ha	diamondback moth, black rot
Carrot	rows 20, thinned to 5 to 8	75 to 110 days after sowing	20 to 30 t/ha	carrot fly, leaf blight

Crop	Spacing (cm)	First harvest	Yield	Watch for
Sweet pepper	60 × 40 to 45	70 to 90 days	about 40 t/ha in the open field	whitefly, thrips, bacterial wilt
Eggplant	75 × 50	75 to 110 days	10 to 12 kg per 10 m row [CRP-70]	fruit borer, bacterial wilt
Amaranth (dodo)	40 × 30	25 to 30 days after sowing, then every week	up to 10 t/ha of leaves [CRP-71]	aphids
Kale (sukuma)	60 × 40 to 60	42 days after transplanting	about 15 t/acre [CRP-69]	diamondback moth, black rot
French beans	30 × 15	42 to 56 days after sowing	bush 5 to 6 t/ha	bean fly, rust
Cucumber	100 × 50	60 days	8 to 10 kg per 10 m row [CRP-70]	aphids, downy mildew
Spinach and chard	45 to 60 × 20 to 25	60 to 90 days	40 to 50 plants per 10 m row [CRP-70]	leaf miners
Herbs	parsley 40 × 15, coriander rows 15	coriander 28 days, parsley 100	small beds near the galley	

Spacing, timing, yields and pests from RAB's vegetable manual [V CRP-68] unless marked.

Before you start

1. **Test the soil.** Take samples from 10 spots across the zone, mix them, and send them to RAB's soil laboratory: pH, organic carbon, nitrogen, phosphorus, texture [V CRP-72].
2. **Test the water.** Measure what the borehole gives over 6 hours, and have a sample tested; the first beds need up to 3.6 m^3 a day in the dry months [E].

Inputs and selling

- **Suppliers** (not contacted; a listing is not stock): Holland Greentech for seed and drip, Davis & Shirtliff for pumps, the Bugesera agro-dealers on RICA's register [V CRP-74].

- **Fertiliser** for vegetables is subsidised through *774# with an ID and the land's UPI; seed is not [V CRP-73].
- **Prices in Bugesera** (farm-gate / retail, RWF per kg, July to October 2026): tomato 350 / 700, onion 475 / 750, cabbage 164 / 350, carrot 550 / 1,000, sweet pepper 850 / 1,500, French beans 550 / 900, amaranth 220 / 400 [V MKT-60]. Kigali pays more: sell the surplus there or to the compound's guests.

Steps

1. **Fence the zone** against animals and keep one gate on the garden walk to the galley.
2. **Mark out the beds:** 1.2 m beds, 0.5 m paths, 1 m paths between blocks of 10 beds; the beds run across the slope.
3. **Build the 60 beds:** dig, mix in compost or manure (about 250 kg per 100 m² for tomatoes [V CRP-68]), raise them 20 cm.
4. **Lay the drip:** a header pipe from the tank, a filter, two drip lines per bed; run it for an hour and check every emitter (water).
5. **Start the nursery:** trays or a seedbed under shade net for kale, cabbage, onion, tomato and sweet pepper.
6. **Sow straight into the beds:** amaranth and French beans, which give a harvest within 4 to 8 weeks.
7. **Add lime only if the test says so:** the vegetables want pH 6.0 to 7.0; the soils here are likely nearer 5.5 to 6.1 [V CRP-68, E LOC-37].

A few beds of each crop go in every few weeks, so something is always ready to pick. Each family moves to another family's beds the next season, so no bed grows the same family twice running.

Crop	Plant	Nursery	To first harvest	Picking
Tomato	2 beds every 4 weeks	5 wk	11 wk	8 wk
Potato	2 beds every 3 weeks	sown in place	13 wk	3 wk
Onion	1 bed every 2 weeks	6 wk	17 wk	2 wk
Cabbage	1 bed every 3 weeks	4 wk	12 wk	3 wk
Carrot	1 bed every 3 weeks	sown in place	13 wk	3 wk
Kale (sukuma)	1 bed every 4 weeks	4 wk	6 wk	16 wk
French beans	3 beds every 4 weeks	sown in place	7 wk	4 wk
Amaranth (dodo)	1 bed every 4 weeks	sown in place	4 wk	6 wk

Crop	Plant	Nursery	To first harvest	Picking
Cucumber	1 bed every 4 weeks	sown in place	8 wk	5 wk
Spinach and chard	1 bed every 4 weeks	4 wk	8 wk	10 wk
Sweet pepper	1 bed every 4 weeks	6 wk	11 wk	10 wk
Eggplant	1 bed every 4 weeks	6 wk	13 wk	10 wk
Herbs	1 bed every 4 weeks	sown in place	6 wk	10 wk
Chilli	1 bed every 4 weeks	6 wk	12 wk	20 wk

Season A · September to February

Raise seedlings for 3 to 8 weeks, transplant with the rains, drip through the January–February lull.

Season B · March to June

Transplant in February and March; the raised beds carry the April rains; drip again from late May.

Season C · July to September

Drip only: plant only what the borehole's proven flow can carry.

Watering

1. The driest months need 4.1 to 5.3 litres per m² of bed a day: 2.8 to 3.6 m³ a day for the first beds [E CRP-75, LOC-54].
2. That is far more than the roofs can collect, so the borehole or the mains carries the field; plant season C only to the flow the test proves.
3. Water at dawn or dusk, never at midday; mulch the beds with dry grass to keep the water in.
4. Under 1 ha of irrigation needs no permit from the Water Resources Board [V LOC-08].

Keeping pests down

1. Rotate: never the same family (tomato and pepper and eggplant; cabbage and kale; beans) in the same bed two seasons running.

2. Walk the beds twice a week; pull and burn any plant that wilts suddenly (bacterial wilt).
3. Hang pheromone traps for Tuta absoluta in the tomato beds.
4. Spray only what the agro-dealer's advisor names for the pest found, and keep a record of every spray.

Hold points

1. **The tests before the beds.** No lime or fertiliser goes on before the soil test is back.
2. **Season C to the proven flow.** Nothing is planted for the dry season beyond what the borehole's test proves it can water.
3. **Every spray recorded.** Only what the advisor names for the pest found, written down.

Done when

1. The soil and water tests are back and the beds are limed and fed to them.
2. The 60 beds are planted, with drip running and no dry emitters.
3. The galley receives a harvest list every morning and the field records every kilogram.

Open questions

1. How much water does the borehole give through June to August, and is it fit for irrigation?
2. Who works the field every day: one gardener, or two?
3. Which Kigali buyers take the surplus, and at what price?
4. A permit for the drip? Under 1 ha needs none [V LOC-08], yet the study's roadmap puts drip after a permit [LOC-04]: which applies?